

AP EAMCET
Engineering Entrance Exam
24th April 2018- Shift 2

Mathematics

1. Match the following

List – I

A) $\frac{x}{e^x - 1} + \frac{x}{2} + 4; x \neq 0$

function

B) $\tan^{-1}(\log(x + \sqrt{x^2 + 1})), x > 0$

C) for $3 < x < 5$

$|x - 2| + |x - 3| + |x - 5|$

D) $\sin 2x + \sin^2 x + \cos 3x \forall x \in \mathbb{R}$

List – II

I) is neither odd even

II) is an even function

III) is an odd function

IV) is the identity function
v) is a constant function

A

B

C

D

A.III

II

V

IV

A

B

C

D

B.III

I

IV

II

A

B

C

D

C.II

III

IV

I

A

B

C

D

D.II

I

V

III

2. If $f(x) = \sqrt{2x-1} + 5 \cos^{-1}\left(\frac{2x-1}{3}\right)$ then the domain of the function $f(x)$ is

A. $\left[-1, \frac{1}{2}\right]$

B. $\left[\frac{1}{2}, 2\right]$

C. $[-1, 2]$

D. $\left[\frac{1}{2}, \infty\right)$

3. $\sum_{n=1}^5 n(n^2 + n + 1) =$

A. 500

B. 155

C. 295

D. 395

4. Let $1, \omega$ and ω^2 be the cube roots of unity. If S is the set of all non-singular matrices of the form

$\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$ where $a, b, c \in \{\omega, \omega^2\}$, then the number of elements in S

is

A. 2

B. 3

C. 4

D. 6

5. If P and Q are two invertible matrices of the same order, then $\text{Adj}(QP)$ is equal to

A. $|P||Q|P^{-1}Q^{-1}$

B. $|P||Q|(PQ)^{-1}$

C. $|P||Q|\text{Adj}(PQ)$

D. $|P||Q|\text{Adj}(QP)$

6. If the equations

$$kx + (k+1)y + (k-1)z = 0$$

$$(k-1)x + (k+2)y + kz = 0 \text{ and}$$

$$(k+1)x + ky + (k+2)z = 0$$

have a non trivial solution, then the sum of all the possible value of k is

A. 0

B. $-\frac{1}{2}$

C. $\frac{1}{2}$

D. 1

7. If ω is a complex cube root of unity and a, b, c are distinct real numbers. Then

$$\frac{a+b\omega+c\omega^2}{c+1\omega+b\omega^2} + \frac{a+b\omega+\omega^2}{b+c\omega+a\omega^2} =$$

A. 1

B. -1

C. $a+b+c$

D. 0

8. If $z = x + iy$ and $z^2 = (i\bar{z})^2$, then

A. $y = \pm \frac{x}{\sqrt{3}}$

B. $y = \pm x$

C. $y = \pm \frac{x}{2}$

D. $x = 0, y = \pm 1$

9. If $z = \frac{1}{1 - \cos \theta + i \sin \theta}$ is acute then the modulus and amplitude of: respectively are

A. $\frac{1}{2} \cos \frac{\theta}{2}, -\left(\frac{\pi}{2} + \frac{\theta}{2}\right)$

B. $\frac{1}{2} \cos \frac{\theta}{2}, \left(\frac{\pi}{2} + \frac{\theta}{2}\right)$

C. $\frac{1}{2} \sec \frac{\theta}{2}, -\left(\frac{\pi}{2} + \frac{\theta}{2}\right)$

D. $\frac{1}{2} \sec \frac{\theta}{2}, \left(\frac{\pi}{2} + \frac{\theta}{2}\right)$

10. If $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are real number $\alpha_1 \neq 0$ and $z = \cos \theta + i \sin \theta$ is a root of the equation $\alpha_1 + \alpha_2 z + \alpha_3 z^2 + \dots + \alpha_n z^{n-1} + z^n = 0$, then $\alpha_1 \cos n\theta + \alpha_2 \cos(n-1)\theta + \dots + \alpha_n \cos \theta =$

A. $1+i$

B. 1

C. -1

D. $1-i$

11. The number of distinct real solution for the equation

$$|x^2 + 2x - 8| + x - 2 = 0$$

A. 1

B. 2

C. 3

D.4

12. If x is real. then the sum of the maximum and the minimum values of the expression $\frac{x^2 + 4x + 1}{x^2 + x + 1}$ is

A.-2

B.2

C.1

D.0

13. If the minimum value of $f(x) = x^2 + 2bx + 2c^2$ is greater than the maximum value of $g(x) = -x^2 - 2cx + b^2$ for all real x then

A. $|c| > \sqrt{2}|b|$

B. $|c|\sqrt{3} > |b|$

C. $-1 < c < \sqrt{2}b$

D. $\frac{1}{\sqrt{2}} < c < |b|$

14. If α, β, γ are the root of $x^2 - 2x^2 + 4x - 1 = 0$ then the equation having the roots $\beta\gamma + \frac{1}{\alpha}, \alpha\beta + \frac{1}{\gamma}, \gamma\alpha + \frac{1}{\beta}$ is

A. $x^3 + 8x^2 - 8x + 8 = 0$

B. $x^3 - 8x^2 + 16x - 8 = 0$

C. $x^3 - 8x^2 + 8x - 8 = 0$

D. $x^3 - 4x^2 + 8x - 16 = 0$

15. The number of ways in which all the letters of the word COMBINATION can be arranged so that the vowels always come together is

A.45678

B.50400

C.75600

D.82300

16. aglasem.com If four points are taken on each of three parallel lines in a plane, then the maximum number of triangles formed with these points is

- A.64
- B.144
- C.208
- D.80

17. The sum of the coefficients of x^{32} and x^{-31} in $\left(x^4 - \frac{1}{x^3}\right)^{15}$

- A.1470
- B.1260
- C.-1260
- D.-1470

18. The coefficient of

x^{15} in the product $(1-x)(1-2x)(1-2^2x)(1-2^3x).....(1-2^{15}x)$ is

- A. $2^{121} - 2^{104}$
- B. $2^{105} + 2^{124}$
- C. $2^{105} - 2^{121}$
- D. $2^{120} - 2^{104}$

19. If $\frac{x^4}{(x-1)^2(x+1)} = Ax + B + \frac{P}{(x-1)} + \frac{Q}{(x-1)^2} + \frac{R}{x+1}$, then $2AP - BQ + R =$

- A.3
- B. $\frac{13}{4}$
- C. $-\frac{11}{4}$
- D. $-\frac{7}{2}$

20. The value of $\tan 40^\circ + \tan 11^\circ + \tan 20^\circ - \tan 56^\circ + \tan 11^\circ + \sqrt{3} \tan 40^\circ \tan 20^\circ$ is

- A. $\sqrt{3} - 1$
- B. $\sqrt{3} + 1$
- C. 1
- D. 0

21. If θ is in the third quadrant, then $\sqrt{4 \sin^4 \theta + \sin^2 2\theta} + 4 \cos^2 \left(\frac{\pi}{4} - \frac{\theta}{2} \right) =$

- A. $1 + 2 \sin \theta$
- B. 2
- C. 1
- D. $2 + 4 \sin \theta$

22. If $\sec(\theta + \alpha), \sec(\theta + \alpha)$ are in arithmetic, then $\cos \theta, \sec \frac{\alpha}{2} =$

- A. $\frac{\sqrt{3}}{2}$
- B. $\frac{1}{\sqrt{2}}$
- C. ± 2
- D. $\pm \sqrt{2}$

23. The number of values of x with $0 \leq x \leq 2\pi$ satisfying the equation $\sin x + \sin 2x + \sin 3x = \cos x + \cos 2x + \cos 3x$ is

- A. 7
- B. 6
- C. 5
- D. 4

24. If $\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{13} + \dots + \tan^{-1} \frac{1}{n^2 + n + 1} = \tan^{-1} \theta$, then $\theta =$

- A. $\frac{n}{n+2}$
- B. $\frac{n}{n+1}$

C. $\frac{n+1}{n+2}$

D. $\frac{n-1}{n+2}$

25. If $u = \log \tan \left(\frac{\pi}{4} + \frac{\theta}{2} \right)$, then $\cosh u =$

A. $\sec \theta$

B. $\tan \theta$

C. $\operatorname{cosec} \theta$

D. $\cot \theta$

26. In $\triangle ABC$, if $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$, then $\cos A \cos B + \cos B \cos C + \cos C \cos A =$

A. -1

B. $\frac{3}{4}$

C. $\frac{9}{4}$

D. 1

27. In $\triangle ABC$, if $A=60^\circ$ and $B=105^\circ$ then

$$\frac{2R^2(b-c)\sin A \sin B \sin C}{(b+c)(s-a \cos C - c \cos A)(s-a \cos B - b \cos A)} =$$

A. $\frac{1}{\sqrt{2}}$

B. $\sqrt{3}$

C. 1

D. $\frac{1}{\sqrt{3}}$

28. In $\triangle ABC$, if $\angle A = 90^\circ$, then $2(r+R) =$

A. $a+b$

B. $b+c$

C. $c+a$

D. $a+b+c$

29. if the position vector of

A, B, C, D are $\vec{i} + 2\vec{j} + 2\vec{k}, 2\vec{i} - \vec{j} + 2\vec{k}, 2\vec{i} - \vec{j} + 3\vec{k}$ and $4\vec{j} + 5\vec{k}$ respectively, then the quadrilateral ABCD is a

A. square

B. Rectangle

C. Rhombus

D. Parallelogram

30. If $(\vec{i} + \vec{j} + \vec{k}), (\vec{i} + 2\vec{j} + 3\vec{k})$ and $(2\vec{i} - \vec{j} + \vec{k})$ are the position vectors of the vertices A, B and C of ΔABC respectively. then the vector equation of the altitude through A is

A. $\vec{r} = \vec{i} + \vec{j} + \vec{k} + t(\vec{i} + 2\vec{j} + 3\vec{k})$

B. $\vec{r} = \vec{i} + \vec{j} + \vec{k} + t(2\vec{i} - \vec{j} + \vec{k})$

C. $\vec{r} = \vec{i} + \vec{j} + \vec{k} + t(\vec{i} - \vec{j} + 2\vec{k})$

D. $\vec{r} = \vec{i} + \vec{j} + \vec{k} + t(4\vec{i} + 2\vec{j} + 4\vec{k})$

31. If the adjacent sides of a rectangle are $\vec{a} = 5\vec{m} - 3\vec{n}, \vec{b} = -\vec{m} - 2\vec{n}$ and the adjacent sides of another rectangle are $\vec{c} = -4\vec{m} - \vec{n}, \vec{d} = \vec{m} + \vec{n}$, then the angle between the vectors $\vec{x} = \frac{\vec{a} + \vec{c} + \vec{b}}{3}$ and $\vec{y} = \frac{\vec{c} + \vec{d}}{5}$ is

A. $\frac{\pi}{2}$

B. $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

C. $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right) + \pi$

D. $\sin^{-1}\frac{19}{5\sqrt{43}}$

32. The perpendicular distance from the point $3\bar{i} - 2\bar{j} + \bar{k}$ to the line joining the points $\bar{i} - 3\bar{j} + 5\bar{k}$ and $2\bar{i} + \bar{j} - 4\bar{k}$ is
- 7
 - $7\sqrt{3}$
 - $2\sqrt{3}$
 - $\sqrt{3}$
33. Let l_1 be the line passing through the point $3\bar{i} + 4\bar{j} + 2\bar{k}$ and parallel to the vector $-\bar{i} + 2\bar{j} + \bar{k}$. Let l_2 be another line passing through the point $\bar{i} - 7\bar{j} - 2\bar{k}$ and parallel to the vector $\bar{i} + 3\bar{j} - 2\bar{k}$. Then the shortest distance between the lines l_1 and l_2 is
- $\sqrt{35}$
 - 9
 - $\sqrt{6}$
 - $\sqrt{29}$
34. If $\bar{c} = 2\bar{i} + \bar{j} - 2\bar{k}$, $\bar{b} = \bar{i} + \bar{j}$ and $\bar{a} = 3\bar{i} + \bar{j} - \bar{k}$, then $(\bar{a} \times (\bar{b} \times \bar{c})) \times \bar{c} =$
- $2\bar{i} - 2\bar{j} + \bar{k}$
 - $18\bar{i} - 18\bar{j} - 9\bar{k}$
 - $-18\bar{i} + 18\bar{j} - 9\bar{k}$
 - $18\bar{i} - 18\bar{j} + 9\bar{k}$
35. The mean of five observations is 4 and their variance is also 4. If three of the five observations are 1, 3, 4 then the product of the other two is
- 32
 - 36
 - 35
 - 27

36. The mean deviation of the following data about its mean is

x_i	5	15	25	35	45
f_i	8	48	56	32	16

A. 25

B. 12

C. 8

D. 16

37. If three distinct vertices are chosen at random from the vertices of a cube. then the probability that they form the vertices of an equilateral triangle. Is

A. $\frac{1}{7}$

B. $\frac{2}{7}$

C. $\frac{1}{2}$

D. $\frac{1}{8}$

38. agiasem.com Let A and B be two independent events such that $P(B) > P(A)$. If the probability that both A and B happen is $\frac{1}{12}$ and the probability that neither A nor B

happens is $\frac{1}{2}$ then

A. $P(A) = \frac{1}{2}, P(B) = \frac{1}{2}$

B. $P(A) = \frac{1}{4}, P(B) = \frac{1}{3}$

C. $P(A) = \frac{1}{6}, P(B) = \frac{1}{5}$

D. $P(A) = \frac{1}{6}, P(B) = \frac{1}{3}$

39. A person is known to speak truth in 3 out of 5 times. If he throws a die and reports that it is six. then the probability that it is actually six. Is
- $\frac{3}{30}$
 - $\frac{13}{30}$
 - $\frac{10}{13}$
 - $\frac{3}{13}$
40. If the probability function of a random variable X is given by $P(X = j) = \frac{1}{2^j} : j = 1, 2, 3, \dots, \infty$ then the variance of X is
- 2
 - 3
 - 4
 - 5
41. If the sum and product of the mean and variance of a binomial distribution are 15 and 54 respectively, then the number of trials in it is
- 27
 - 54
 - 28
 - 81
42. A(5, 3). B(3, - 2). C(2, -1) are three points. If P is a variable point such that the area of the quadrilateral PABC is 10sq. units, then the locus of P is
- $16x^2 - 24xy + 9y^2 - 144x + 108y - 76 = 0$

B. $16x^2 - 24xy + 9y^2 + 144x - 108y - 76 = 0$

C. $16x^2 - 24xy + 9y^2 - 144x + 108y + 76 = 0$

D. $16x^2 - 24xy + 9y^2 + 144x - 108y + 76 = 0$

43. If the origin is shifted to the point (1, 1) and the axes are rotated through an angle 45° about this point, then the transformed equation of the equation $x^2 + 2xy + y^2 - 1 = 0$ is

A. $2y^2 - 4\sqrt{2}y - 3 = 0$

B. $2y^2 + 4\sqrt{2}y - 3 = 0$

C. $2x^2 + 4\sqrt{2}x + 3 = 0$

D. $2x^2 - 4\sqrt{2}y + 3 = 0$

44. A line is passing through the point (4, 3) and the sum of its intercepts made on the coordinate axes is 14. Then an equation of that line is

A. $3x + 4y = 24$

B. $6x + 8y = 24$

C. $4x + 3y = 15$

D. $4x - 3y = 7$

45. If (x_1, y_1) and (x_2, y_2) are two points on the line $x + y + 3 = 0$ such that each of them is at a distance of $\sqrt{5}$ units from the line $x + 2y + 2 = 0$, then $x_1 + x_2 + y_1 + y_2 =$

A. -6

B. 20

C. 6

D. -20

46. Match the following,

List-I

A) The equation of line passing through (4,3) whose

B) The equation of the line passing through the centroid and circumcenter of $\triangle ABC$ with vertices

A (1,1), B (3,3), C (6, -6)

C) The equation of the line whose X – intercept is $(-\frac{3}{5})$

And is perpendicular to $x - y + 2 = 0$

D) The equation of the line whose distance from the origin is 2 and the normal drawn from the origin makes an angle 45° with the positive direction of X-axis

List- II

I) $x + y - 2\sqrt{2} = 0$

II) $7x + 23y - 8 = 0$

III) $x + 2y + \sqrt{2} = 0$

IV) $x + 2y - 10 = 0$

V) $5x + 5y + 3$

A B C D

A. V IV II I

A B C D

B. III V IV II

A B C D

C. IV II V I

A B C D

D. II I III V

47. If (a, b) is the centroid of the triangle $4x^2 - 17xy + 4y^2 = 0$ and $x + y - 5 = 0$ and c is numerical value of the area of the triangle, then $a + b + c =$

A. $\frac{25}{6}$

B. $\frac{25}{3}$

C. $\frac{65}{6}$

D. $\frac{15}{3}$

48. If the lines represented by the equation $4xy + 6x - 8y + c = 0$ form a rectangle with the coordinate axis then the area of the rectangle (in sq. units) is

A. $|c|$

B. $\left|\frac{c}{2}\right|$

C. $\left|\frac{c}{4}\right|$

D. $\left|\frac{c}{8}\right|$

49. The sum of the minimum and maximum distances of the point (4, -3) to the circle $x^2 + y^2 + 4x - 10y - 7 = 0$ is

A. 20

B. 16

C. 12

D. 64

50. If the tangent to the circle $x^2 + y^2 - 4x + 2y - 5 = 0$ at $(3, -4)$ cuts the circle $x^2 + y^2 + 16x + 2y + 10 = 0$ at A and B. then the mid point of AB is:
- A. $(-6, -9)$
 - B. $(-9, -6)$
 - C. $(-6, -7)$
 - D. $(-7, -6)$
51. If the circle $x^2 + y^2 + 8x - 4y + c = 0$ touches the circle $x^2 + y^2 + 2x + 4y - 11 = 0$ externally and cuts the circle $x^2 + y^2 - 6x + 8y + k = 0$ orthogonally. then $k =$
- A. -59
 - B. -19
 - C. 19
 - D. 59
52. If a system of circles pass through $(2, 3)$ and cuts the circle $x^2 + y^2 = 12$ orthogonally, then the equation of the locus of the centers of that system of circles, is
- A. $4x + 6y - 13 = 0$
 - B. $4x + 6y - 25 = 0$
 - C. $6x + 4y - 25 = 0$
 - D. $6x + 4y - 13 = 0$
53. The radius of the circle whose diameter is the common chord of the circles $x^2 + y^2 + 2x + 2y + 1 = 0$ and $x^2 + y^2 + 4x + 3y + 2 = 0$ is
- A. $\frac{2}{\sqrt{5}}$
 - B. $\frac{1}{\sqrt{5}}$
 - C. 1

D. $\frac{\sqrt{17}}{2}$

54. If a focal chord of the parabola $y^2 = 4x$ makes an angle 45° with positive X-axis, then the slopes of the normals drawn at the ends of the focal chord will satisfy the equation

A. $x^2 - 2x - 1 = 0$

B. $x^2 + 2x - 1 = 0$

C. $x^2 - 1 = 0$

D. $x^2 + 2x - 2 = 0$

55. If the angle between the tangents drawn through the point $(-2, -1)$ to the parabola $y^2 = 4x$ is θ . then $\tan 2\theta =$

A. 3

B. -3

C. $-\frac{3}{4}$

D. $\frac{3}{4}$

56. If α, β are the eccentric angles of the extremities of a focal chord (other than the major axis) of the ellipse

$$x^2 + 4y^2 = 4 \text{ then } \sqrt{3} \cos \frac{\alpha + \beta}{2} =$$

A. $2 \cos \frac{\alpha + \beta}{2}$

B. $2 \sin \frac{\alpha - \beta}{2}$

C. $2 \sec \frac{\alpha + \beta}{2}$

D. $2 \sin \frac{\alpha + \beta}{2}$

57. The equations of the tangents to the ellipse $9x^2 + 16y^2 = 144$ which pass through the point $(2, 3)$ are

A. $x + y = 5, y = 3$

B. $x + y + 5 = 0, y = 3$

C. $x + y = 5, y + 3 = 0$

D. $x + y + 5 = 0, y + 3 = 0$

58. The locus of the point of intersection of the tangents drawn at the extremities of a normal chord of the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is}$$

A. $\frac{a^2}{x^2} - \frac{b^2}{y^2} = a^2 + b^2$

B. $\frac{a^4}{x^2} - \frac{b^4}{y^2} = (a^2 - b^2)^2$

C. $\frac{a^3}{x^2} - \frac{b^3}{y^2} = (a^2 + b^2)^2$

D. $\frac{a^6}{x^2} - \frac{b^6}{y^2} = (a^2 + b^2)^2$

59. For $a \neq 0$. if the sum of the distances of a point from the points $(a, 0, 0)$ and $(-a, 0, 0)$ is a constant $2k$. then the locus of that point is

A. $x^2 + k^2(y^2 + z^2) = k^2$

B. $\frac{x^2}{k^2} + \frac{y^2 + z^2}{k^2 - a^2} = 1$

C. $\frac{x^2}{k^2 - a^2} + \frac{y^2 + z^2}{k^2} = 1$

D. $x^2 + y^2 + z^2 = \frac{1}{k^2 + 1}$

60. If $P = (2, -3, 4)$, $Q = (-1, -4, 0)$, $R = (2, 1, 0)$ are three points and S is the foot of the perpendicular drawn from R to PQ, then the X-coordinate of S is

A. $\frac{-45}{13}$

B. $\frac{28}{13}$

C. $\frac{6}{7}$

D. $\frac{8}{13}$

61. π is a plane passing through the origin and containing two lines whose direction ratios are 1, -2, 2 and 2, 3, -1 then direction ratios of the line of intersection of the planes $x - y - z + 1 = 0$ and π

are

A. 2, 1, 1

B. 2, 3, -1

C. 1, -2, 2

D. -1, -1, 0

62. If α and β are the roots of the equation $ax^2 + bx + c = 0$.

then $\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2} =$

A. $\frac{a^2(\alpha - \beta)^2}{4}$

B. 1

C. $\frac{a(\alpha - \beta)^2}{2}$

D. $\frac{a^2(\alpha - \beta)^2}{2}$

63. $\lim_{x \rightarrow 0} \left(\frac{4^x - 1}{2^x - 1} - \frac{\sqrt{4 + 3x} - 2}{x} \right) =$

A. 0

B. $\frac{5}{4}$

C. $\log 2 - 3$

D. $\frac{\log 4}{\log 2} - 3$

64. Define the function $f: \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} (1+3x)^{\frac{4}{x}}, & \text{if } x \neq 0, \\ a & \text{if } x = 0. \end{cases}$$

If f is continuous, then $\log a =$

A. $\frac{1}{3}$

B. $\frac{3}{4}$

C. $\frac{4}{3}$

D. 12

65. If $a > b > 0$ and x is acute, then $\frac{d}{dx} \left[\cos^{-1} \left(\frac{b - a \cos x}{a - b \cos x} \right) \right] =$

A. $\frac{a^2 - b^2}{b \cos x - a}$

B. $\frac{a^2 - b^2}{a - b \cos x}$

C. $\frac{\sqrt{a^2 - b^2}}{a - b \cos x}$

D. $\frac{\sqrt{a^2 - b^2}}{b \cos x - a}$

66. Match the following:

List – I

$f(x, y) = 0$

A) $x^2 + y^2 + 3xy = 7$

B) $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$

C) $x^3 + y^3 = 3axy$

D) $xy(x + y) = 2$

List – II

$\frac{dy}{dx}$

I) $\frac{x^2 + ay}{ax + y^2}$

II) $\frac{-(2x + 3y)}{3x + 2y}$

III) $-\left(\frac{y}{x}\right)^{\frac{1}{3}}$

IV) $\frac{x^2 - ay}{ax - y^2}$

$$V) \frac{-y(2x+y)}{x(x+2y)}$$

The correct answer is

- | | | | | |
|----|----|-----|----|----|
| | A | B | C | D |
| A. | II | III | IV | I |
| | A | B | C | D |
| B. | II | III | I | IV |
| | A | B | C | D |
| C. | II | III | IV | V |
| | A | B | C | D |
| D. | II | III | V | IV |

67. If $y^2 = ax^2 + 2x + c$, then $y^3 \frac{d^2y}{dx^2}$ is

- A. $ac - 1$
 B. $ac + 1$
 C. $4ac - 1$
 D. $ac - 4$

68. If $f(x): \mathbb{R} \rightarrow \mathbb{R}$ be a function defined for all $x \in \mathbb{R}$ by

$f(x) = x^2 + f'(1)x^2 + f''(2)x - f'''(3)$ then the area (in sq. units) of the triangle formed by X-axis, the tangent and the normal drawn to the curve $y = f(x)$ at $x = 0$ is

- A. 90
 B. 180
 C. 45
 D. 130

69. The set of values of x for which $f(x) = 3x^4 - 8x^3 - 6x^2 + 24x - 12$ is an increasing function, is

- A. $(-\infty, -1) \cup (1, 2)$

B. $(-1, 2)$

C. $(-1, 1) \cup (2, \infty)$

D. $\mathbb{R}$

70. If a function is differentiable on $\mathbb{R}$ such that

$f'(x) \leq 4$ for all $x \in \mathbb{R}$; and if $f(2) = -6$ and $f(6) = 8$, then the value of $f(4)$

belongs to the interval

A. $[-2, -1]$

B. $[3, 4]$

C. $[0, 2]$

D. $[5, 8]$

71. If $f(x) = 2x^6 - 3$ then

A. the local minimum value of f is zero

B. -3 is the local maximum value of f

C. -3 is the local minimum value of f

D. f has neither minimum nor maximum

72. If $\int \frac{x^{\frac{1}{2}}}{\sqrt{a^3 - x^3}} dx = P(x) + c$

A. $\frac{1}{3} \sin^{-1} \left(\frac{x^3}{a^3} \right)$

B. $\frac{2}{3} \cos^{-1} \left(\frac{x}{a} \right)$

C. $\frac{2}{3} \sin^{-1} \left(\frac{x}{a} \right)^{\frac{3}{2}}$

D. $\frac{1}{2} \sin^{-1} \left(\frac{x}{a} \right)^{\frac{1}{2}}$

73. $\int \tan^{-1}\left(x^{\frac{1}{3}}\right) dx =$

A. $\frac{1}{2} \log\left(1+x^{\frac{2}{3}}\right) - \frac{1}{2} x^{\frac{2}{3}} + c$

B. $x \tan^{-1}\left(x^{\frac{1}{3}}\right) - \frac{1}{2} x^{\frac{2}{3}} + c$

C. $\frac{1}{2} \log\left(1+x^{\frac{1}{3}}\right) - \frac{1}{2} x^{\frac{2}{3}} + c$

D. $x \tan^{-1}\left(x^{\frac{1}{3}}\right) + \frac{1}{2} \log\left(1+x^{\frac{2}{3}}\right) - \frac{1}{2} x^{\frac{2}{3}} + c$

74. If $\frac{3x+1}{(x-1)^3(x+1)} dx = A \log\left|\frac{x+1}{x-1}\right| + \frac{B}{x-1} + \frac{C}{(x-1)^2} + D$ then $A+B+C =$

A. $-\frac{5}{4}$

B. $\frac{5}{4}$

C. $-\frac{5}{2}$

D. $\frac{5}{2}$

75. For any integer $n \geq 2$, if $I_n = \int \cot^n x dx$ then $I_5 =$

A. $-\frac{\cot^4 x}{4} + \frac{\cot^2 x}{2} + \log|\sin x| + c$

B. $-\frac{\cot^4 x}{4} + \frac{\cot^2 x}{2} - \log|\sin x| + c$

C. $\frac{\cot^4 x}{4} + \frac{\cot^2 x}{2} + \log|\cos x| + c$

D. $\frac{\cot^4 x}{4} + \frac{\cot^2 x}{2} - \cot x + c$

76. $\lim_{n \rightarrow \infty} \sum_{n=5}^{5n+5} \frac{n^2}{(n^2+r^2)^{\frac{3}{2}}} =$

A. $\frac{1}{\sqrt{2}}$

B. $\frac{1}{3} \log 2$

C. $\frac{\pi}{4}$

D. $\frac{5}{\sqrt{26}}$

77. If $a_n = \int_0^{\frac{\pi}{2}} \frac{\sin^2 nx}{\sin x} dx$, then $a_2 - a_1, a_3 - a_2, a_4 - a_3$ are in

A. Arithmetic progression

B. Geometric progression

C. Harmonic progression .

D. Aritlametico-geometric progression

78. The area (in sq. units) of the region bounded by the curves

$y = x \log x$ and $y = 2x - 2x^2$ is

A. $\frac{1}{12}$

B. $\frac{7}{6}$

C. $\frac{7}{3}$

D. $\frac{7}{12}$

79. The differential equation of the family of curves $r^2 = a^2 \cos 2\theta$ where 'a' is an arbitrary constant is:

A. $\frac{dr}{d\theta} = r \cot 2\theta$

B. $\frac{dr}{d\theta} = r \tan 2\theta$

C. $\frac{dr}{d\theta} = r \sin 2\theta$

D. $\cos 2\theta \cdot \frac{dr}{d\theta} + r \sin 2\theta = 0$

80. The general solution of the differential equation $\frac{dy}{dx} = \frac{x+2y-3}{2x+y-3}$

is

A. $x = 2(x-y) + \log(x-y+2) + c$

B. $\left(x+y+\frac{4}{3}\right)(x-y+2)^3 = c^2$

C. $(2x-y)^2 = c(x+2y-3)$

D. $x+y-2 = c^2(x-y)^3$